Hierarchical VAE for Chest X-ray Synthesis

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Abstract

We implement a hierarchical convolutional VAE for 128×128 chest radiographs and evaluate independent prior samples, not merely reconstructions. Weak adversarial decoder refinement improves PyTorch-Ignite FID modestly over three seeds, with no corresponding Inception Score gain. Matched reconstructions and generated images highlight this distinction.

1 Method

Data and protocol. The assigned Chest X-Ray Pneumonia collection (Mooney, 2018) is resized to 128×128 grayscale. A seeded split yields 4,694 training and 522 validation images; the separate test set is untouched. Synthesis is unconditional (no labels or input images).

Hierarchical architecture. The original PyTorch H3B model (7.20M parameters) uses residual convolutional blocks and GroupNorm, with a 128-D global latent z_g and an $8 \times 32 \times 32$ local Gaussian latent z_ℓ (Kingma and Welling, 2014; Sønderby et al., 2016; Vahdat and Kautz, 2020). The joint model and inference distribution factorize as

$$\begin{aligned} p_\theta(x, z_g, z_\ell) &= p(z_g)p_\theta(z_\ell \mid z_g)p_\theta(x \mid z_g, z_\ell), \\ q_\phi(z_g, z_\ell \mid x) &= q_\phi(z_g \mid x)q_\phi(z_\ell \mid x, z_g), \\ p(z_g) &= \mathcal{N}(0, I). \end{aligned} \tag{1}$$

We reparameterize each Gaussian as $z = \mu + \exp(\frac{1}{2}\log\sigma^2)\epsilon$; the local posterior corrects the conditional prior. Independent generation samples $z_g \sim p(z_g)$ then $z_\ell \sim p_\theta(z_\ell \mid z_g)$, with no image-to-decoder skip connections.

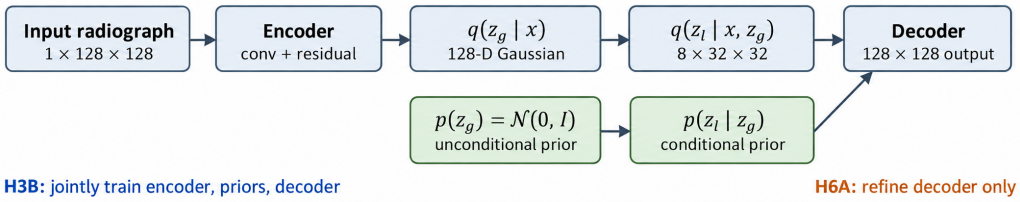


Figure 1: H3B inference (top) and prior sampling (bottom); H6A refines the decoder.

Objective and optimization. We use a reconstruction-regularized VAE objective rather than a strict negative ELBO:

$$\mathcal{L} = \mathrm{MSE}_{\mathrm{sum}} + 48(1 - \mathrm{SSIM}) + 32\mathcal{L}_{\mathrm{Sobel}} + \beta_t\big(\mathrm{KL}_g + 0.35\mathrm{KL}_\ell\big). \tag{2}$$

Here $\mathcal{L}_{\mathrm{Sobel}}$ is Sobel edge-map L_1 error and SSIM is structural similarity (Wang et al., 2004). KL warm-up increases β_t from 0.1 to 1 over 10 epochs. H3B uses AdamW (LR 1.5×10^{-4}, weight decay 10^{-4}), batch 16, 50 epochs, gradient clipping 1 and decoder EMA 0.999; selection uses validation loss.

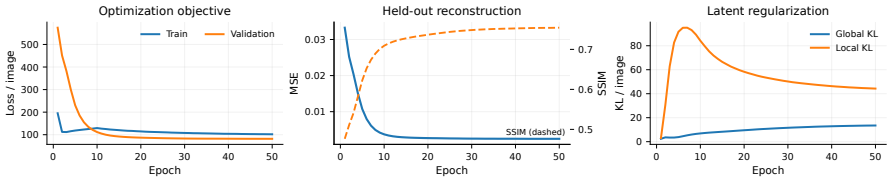


Figure 2: Fifty-epoch H3B training: objective, held-out MSE/SSIM, and global/local KL.

2 Experiments, results, and discussion

Decoder-only refinement. H6A retains H3B's frozen encoder and global/local priors and fine-tunes 2.93M decoder parameters for 15 epochs with a spectral-normalized PatchGAN (Miyato et al., 2018). Real images, posterior reconstructions and independent prior samples train the critic. Adaptive hinge supervision targets a 5% adversarial-to-reconstruction final-layer gradient ratio (decoder/critic LR $10^{-5}/5 \times 10^{-5}$). H6A is an auxiliary refinement, not a second pure-VAE architecture.

FID/IS evaluation. Following the assigned PyTorch-Ignite tutorial (PyTorch-Ignite, 2021; Salimans et al., 2016), we evaluate pretrained InceptionV3 `ignite.metrics.FID` and `InceptionScore`. For each seed 2026, 2027 and 2028, we compare 522 *independent prior draws* per model with the same 522 held-out radiographs. All grayscale real/fake images receive identical 8-bit RGB conversion and 299×299 PIL-bilinear resizing. The tutorial-matched default FID uses 1,000-D classifier logits rather than pooled 2,048-D features; their numerical scales are not interchangeable (Heusel et al., 2017; Parmar et al., 2022).

Table 1: Ignite FID ↓ and IS ↑ (522 real and generated images per seed; means ± sample SD across three sampling seeds).

Model	FID '26	FID '27	FID '28	FID mean±SD	IS mean±SD
H3B (VAE)	.14163	.14418	.16113	.14898±.01060	**1.7353±.0139**
H6A (refined)	**.13574**	**.13932**	**.15348**	**.14285±.00938**	1.7336±.0149

Quantitative interpretation. H6A lowers Ignite FID for all three seeds (mean $0.14898 \rightarrow 0.14285$; -4.12%), while mean IS remains nearly unchanged. A *separate*, pooled-2048 evaluation at seed 2026 gives FID $214.81 \rightarrow 208.996$ and KID $0.2702 \rightarrow 0.2613$ (H3B→H6A). The Ignite train–validation real–real FID is 0.00404 on seed 2026, a finite-sample reference, not a target score. Three seeds against one fixed validation set do not establish statistical significance.

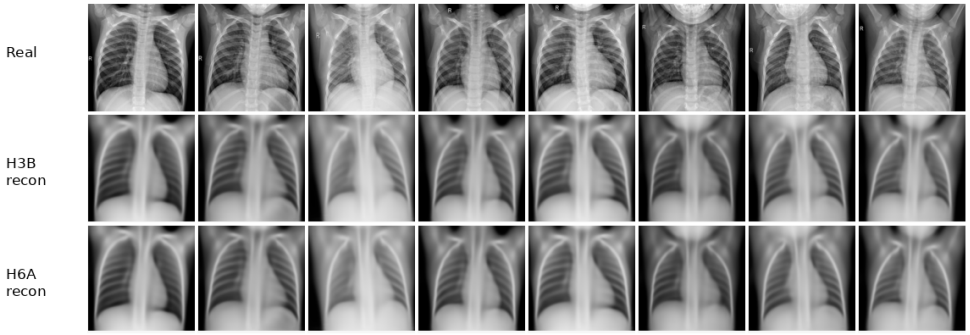


Figure 3: Eight held-out cases: real radiographs (top), H3B posterior reconstruction (middle), and H6A reconstruction (bottom). Reconstructions are *not* novel samples.

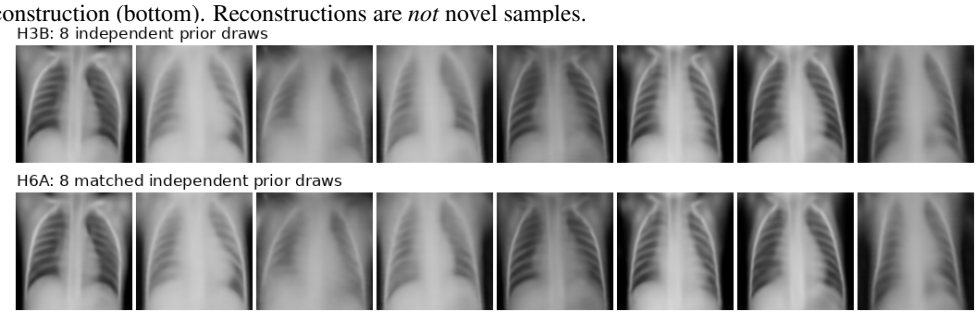


Figure 4: Matched *unconditional* prior samples (8 H3B, 8 H6A; seed 2028). The source package includes 32-image galleries.

Limitations. H3B reaches validation MSE ≈ 0.00248 and SSIM ≈ 0.754. On 16 matched reconstructions, H3B/H6A MSE is 0.002556/0.002564, so improved generation FID is not attributable to reconstruction gains. Global thoracic shapes are recognizable, but ribs and pulmonary markings remain blurred; ImageNet FID is not clinical validation and is sensitive to resampling (Parmar et al., 2022). These results demonstrate reproducible prior-based generation, not diagnostic-quality synthesis.

References

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